

# Posterior-Based Decision Stability in Urban Flood Risk Prioritisation

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## Abstract

This project investigates the stability of prioritisation decisions in urban pluvial flood risk assessments under epistemic uncertainty. Existing approaches often quantify uncertainty within individual components of the exposure–hazard–impact chain, but rarely propagate it coherently to the prioritisation outputs that guide intervention.

The project proposes a unified hierarchical Bayesian framework linking exposure, hazard, and impact within a single generative model. Rather than focusing only on expected loss estimates, the framework derives posterior distributions over prioritisation rankings. Decision-level quantities such as rank distributions and top-(k) membership probabilities are treated as primary inferential outputs.

The empirical reference case is Rotterdam, where a city-scale spatial pipeline has already been developed for approximately 221,000 buildings. The framework is further extended through cross-city transfer under a fixed generative specification, interpreted as a structural stress test.

The project reframes urban pluvial risk modelling from expected-loss estimation toward inference about the robustness of prioritisation decisions under uncertainty.

## 1. Introduction and Problem

Urban flood risk assessments are routinely used to support infrastructure investment and resilience planning. In practice, these decisions often take the form of prioritisation: given limited resources, which buildings, streets, or neighbourhoods should be protected first?

A municipality may need to select a small subset of assets, for example the 500

most critical buildings out of more than 200,000, for targeted mitigation. This selection is usually based on rankings derived from estimated risk or expected loss.

The difficulty is that such rankings are built on uncertain inputs. Exposure, hazard, and impact are all modelled under incomplete data, imperfect assumptions, and epistemic uncertainty. Yet in many practical workflows this uncertainty is reduced to point estimates and then converted into a single deterministic ranking.

In practice, this usually means estimating a risk value for each asset and then sorting all assets from highest to lowest priority. In simplified form, that decision step can be written as

$$\text{rank}_i = \text{argsort}(\mathbb{E}[\text{risk}_i])$$

This creates a mismatch between modelling and decision-making. Uncertainty is acknowledged within components of the model, but it disappears at the very step where intervention priorities are derived. As a result, it remains unclear whether prioritisation outcomes are robust or whether small changes in assumptions could lead to different decisions.

This project addresses that gap by treating prioritisation itself as an uncertain quantity. Instead of producing one ranking, it derives a distribution over rankings from a joint posterior distribution.

The central research question is therefore:

**How stable are prioritisation decisions in urban pluvial risk assessments once uncertainty is propagated coherently across the full exposure–hazard–impact chain?**

In simple terms: if two assets are ranked near the decision boundary, are we looking at a robust priority or at a hidden coin toss?

## 2. Literature Review and Positioning

Research on uncertainty in flood risk assessment is substantial, but it is spread across several partially disconnected strands.

A first strand consists of **Bayesian damage models**, such as Sairam et al. (2019), Mohor et al. (2021), and Lv et al. (2021). These studies show that hierarchical Bayesian approaches are effective for modelling flood damage under sparse and heterogeneous data conditions. However, their inferential targets remain parameter estimation, predictive performance, and transferability of impact models. Prioritisation is still usually derived from expected losses or summary statistics.

A second strand focuses on **end-to-end physical uncertainty propagation**, for example McMillan and Brasington (2008). This work recognises that uncertainty accumulates across hydrological and hydraulic model cascades. Yet the main focus remains physical process representation, and the final prioritisation step is typically performed outside the probabilistic framework.

A third strand uses **Bayesian networks and susceptibility models**, such as Wu et al. (2019, 2020), to produce probabilistic risk maps from heterogeneous variables. These approaches are useful for integrating multiple factors, but they do not derive posterior distributions over asset-level prioritisation rankings within a hierarchical exposure–hazard–impact structure.

A fourth strand includes **robust decision-making under deep uncertainty**, such as Info-Gap and RDM (Hall & Harvey, 2009; Matrosov et al., 2013). This literature is highly relevant because it addresses robustness explicitly. However, it generally operates on scenario ensembles and policy alternatives rather than on posterior distributions over ranking-based prioritisation outputs.

Taken together, these strands show that uncertainty is already recognised as important in flood risk assessment. What remains missing is not awareness of uncertainty itself, but a framework that carries it coherently into the ranking outputs that directly support intervention decisions. Across these strands, uncertainty is recognised, but the final decision output is still rarely treated as the primary object of probabilistic inference.

The present project sits at the intersection of these strands but makes a different move. It does not propose a new hydraulic model, a new damage function, or a new planning paradigm. Its contribution is to connect exposure, hazard, and impact within a single probabilistic framework and to treat prioritisation stability itself as the inferential target.

### 3. Research Gap and Contribution

The central gap lies at the interface between probabilistic modelling and decision-making.

Existing work demonstrates how uncertainty can be represented: - within model components, - across physical model chains, - or across planning scenarios.

What remains largely missing is a framework that carries epistemic uncertainty coherently into the final prioritisation outputs used for intervention.

This project addresses that gap by redefining the inferential target. Rather than treating expected loss as the main output, it treats prioritisation decisions themselves as posterior quantities.

The main novelty therefore lies not in proposing a new hazard model, but in shifting the inferential focus from modelled risk to the stability of the prioritisation decisions derived from it.

The project makes three main contributions:

1. **Unified probabilistic structure**  
It formalises exposure, hazard, and impact within a single hierarchical posterior.
2. **Decision as a posterior quantity**  
It derives rankings and top-(k) membership from posterior draws rather than from expected values alone.
3. **Transfer as structural stress test**  
It evaluates the same generative specification across cities and interprets changes in ranking stability as evidence of structural robustness or sensitivity.

In short, the project is not about building a better flood map. It is about determining how reliable the decisions based on such models actually are.

## 4. Methodological Framework

The framework conceptualises urban pluvial risk as a probabilistic system linking exposure, hazard, impact, and decision:

$$\text{Exposure} \rightarrow \text{Hazard} \rightarrow \text{Impact} \rightarrow \text{Decision}$$

At the asset level, the relevant quantities are: - exposure indicators ( $X\_i$ ), - hazard intensity ( $H\_i$ ), - impact outcome ( $Y\_i$ ).

The generative structure is

$$H_i \sim p(H \mid X_i, \theta_H)$$

$$Y_i \sim p(Y \mid H_i, X_i, \theta_Y)$$

where ( $\_H$ ) and ( $\_Y$ ) denote hazard and impact parameters.

Hazard is treated as a probabilistic input variable informed by long-term precipitation patterns rather than as a fully resolved hydrodynamic simulation. In the current empirical implementation, ( $H\_i$ ) is approximated by an observed precipitation-derived proxy. More complex latent hazard modelling is reserved for later stages of the project.

Bayesian inference yields a joint posterior over plausible system states:

$$p(\theta_H, \theta_Y, H \mid \text{data}) \propto p(Y \mid H, X, \theta_Y) p(H \mid X, \theta_H) p(\theta_H) p(\theta_Y)$$

This is important because the same inferential structure generates both predictive quantities and decision-level outputs.

In Phase 1, the impact model is specified as a simple logistic regression with synthetic binary outcome:

$$\text{logit}(p_i) = \alpha + \beta_E X_i + \beta_H H_i$$

$$Y_i \sim \text{Bernoulli}(p_i)$$

with weakly informative priors:

$$\alpha, \beta_E, \beta_H \sim \mathcal{N}(0, 2.5)$$

This baseline model is intentionally modest. Its purpose is not hydraulic completeness, but transparent end-to-end uncertainty propagation.

Figure 1 — Hierarchical Exposure-Hazard-Impact Model (general formulation)

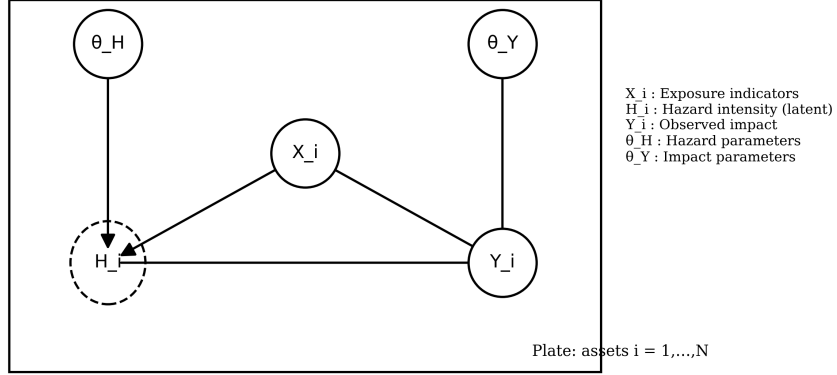


Figure 1: Hierarchical exposure-hazard-impact model

## 5. Decision-Level Inference

The central methodological shift occurs at the final step.

In conventional workflows, risk is estimated and then collapsed into a single ranking. In the present framework, for each posterior draw  $(s)$ , asset-level risks are computed and converted into rankings:

$$\text{rank}^{(s)} = \text{argsort}(\text{risk}^{(s)})$$

This means prioritisation is represented not as one ordering, but as a distribution over orderings.

The main decision-level quantities are:

- posterior rank distributions ( $\hat{r}_i^{(s)}$ ),
- top-(k) membership probabilities,

$$P(i \in \text{top-}k) = \frac{1}{S} \sum_{s=1}^S \mathbf{1}(i \in \text{top-}k^{(s)})$$

- and ranking variability measures such as rank standard deviation.

These quantities allow a distinction between robust and unstable priorities. If an asset appears in the top-(k) group in nearly every posterior draw, its priority is stable. If it moves in and out of the group depending on the draw, then it lies near a fragile decision boundary.

In plain terms: instead of saying “this building is ranked fifth,” the framework allows us to say “this building is almost always among the highest priorities” or “this building is only sometimes there.” That difference is exactly what matters when budgets force sharp decisions.

## 6. Empirical Basis and Current Progress

The empirical reference case is the municipality of Rotterdam, defined using the CBS gebiedsindelingen 2025 administrative boundary and comprising approximately 221,000 building footprints after clipping the raw OpenStreetMap regional extract. A city-scale spatial data integration pipeline has already been developed prior to the present phase of modelling work.

The current dataset integrates: - building footprints from OpenStreetMap regional extracts (via Geofabrik), - the authoritative Rotterdam municipality boundary from CBS gebiedsindelingen 2025 (`gemeente_niet_gegeneraliseerd`), used as the spatial clip, - hydrography layers, - ERA5-Land precipitation data, - exposure indicators such as distance to water bodies and local hydrographic density.

This work is important because the project does not begin from an abstract idea alone. It already rests on an operational asset-level pipeline at city scale.

The completed preparatory work includes: - reproducible spatial preprocessing, - harmonisation of geospatial sources and identifiers, - generation of building-level exposure indicators, - construction of a precipitation-based hazard proxy, - implementation of an initial Bayesian baseline model, - computation of preliminary decision-stability metrics.

The current implementation uses a synthetic binary outcome variable and a simplified hazard proxy. These simplifications are intentional. They allow the

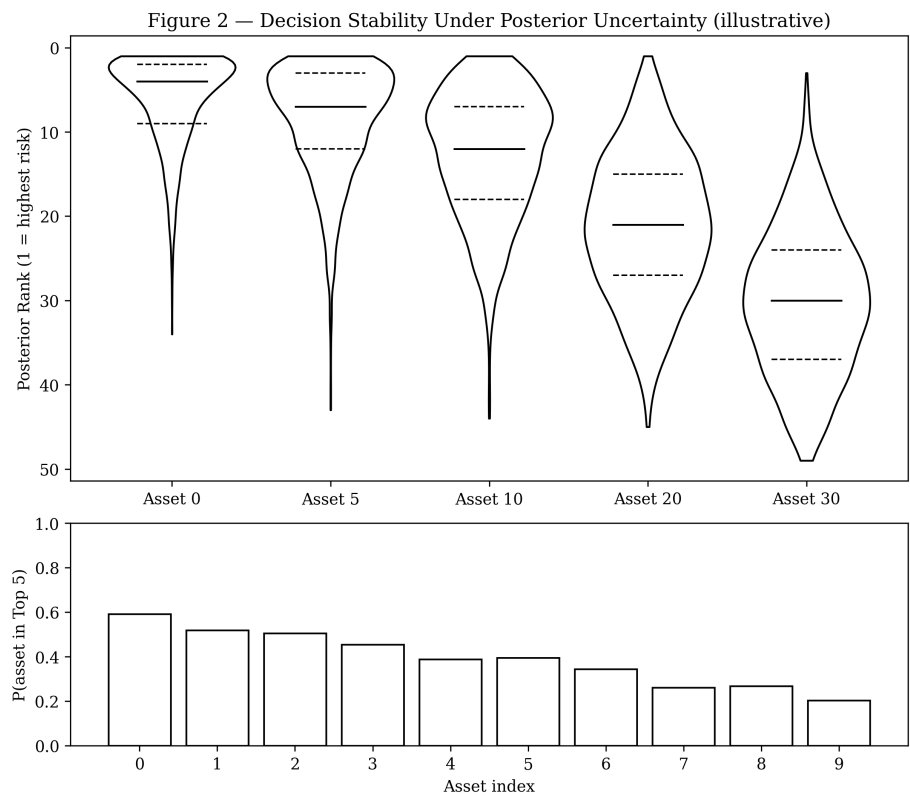


Figure 2: Decision stability concept

project to validate the probabilistic inference and decision pipeline before moving to richer hazard formulations and cross-city transfer.

A first citywide output of the Phase 1 model is the distribution of posterior mean impact probabilities across all buildings.

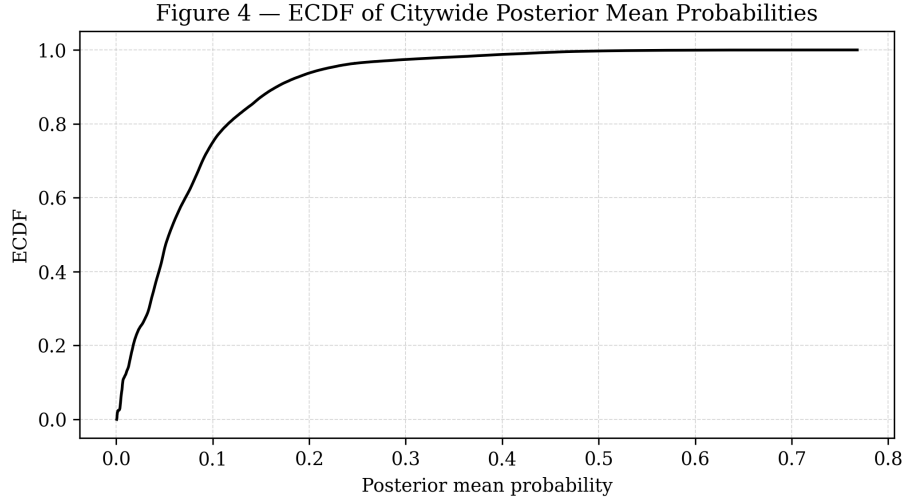


Figure 3: Empirical cumulative distribution of citywide posterior mean probabilities

Preliminary results already indicate that decision instability is concentrated near relatively narrow prioritisation boundaries rather than spread uniformly across the urban asset set.

## 7. Cross-City Transfer and Work Plan

A key part of the project is to apply the same model structure to additional cities without redefining the generative specification.

This is not treated as recalibration. It is treated as a **structural stress test**.

If ranking behaviour remains stable across contexts, that suggests the framework captures relationships that generalise. If ranking variability expands under transfer, that indicates sensitivity to contextual change and therefore reveals limits of the modelling structure.

The project is organised in four phases from March 2026 to February 2027:

### Phase 1 — Baseline probabilistic pipeline (completed)

- finalise Rotterdam dataset,

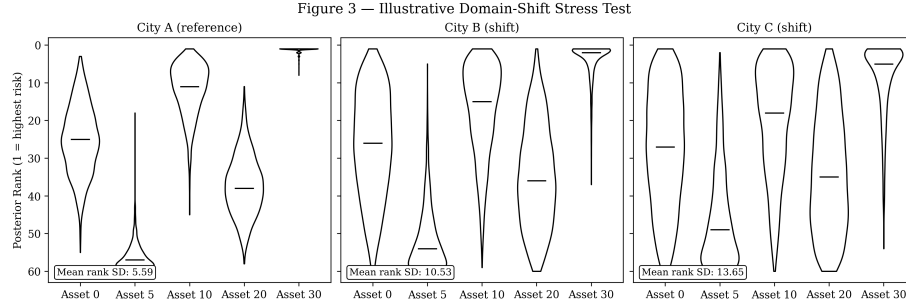


Figure 4: Domain-shift stress test concept

- implement baseline Bayesian model,
- compute posterior decision metrics,
- prepare methodological working paper.

## Phase 2 — Posterior decision stability analysis

- extend posterior analysis,
- test prior and specification sensitivity,
- evaluate alternative hazard proxies,
- refine ranking stability measures.

## Phase 3 — Cross-city structural transfer

- apply the fixed-specification model to additional cities,
- analyse ranking behaviour under domain shift,
- quantify instability changes across contexts.

## Phase 4 — Model extension and synthesis

- introduce a latent hazard layer,
- compare proxy-based and latent hazard formulations,
- integrate findings across cities,
- prepare journal manuscript.

# 8. Expected Contributions and Scope

The project is primarily methodological, with empirical work serving to demonstrate the framework in practice.

Its main contributions are:

- formalisation of the exposure–hazard–impact chain within a unified hierarchical posterior;

- treatment of prioritisation stability as an inferential target;
- derivation of ranking distributions and top-(k) membership probabilities from posterior draws;
- use of cross-city transfer as a diagnostic of structural robustness.

Empirically, the project is expected to: - quantify how epistemic uncertainty affects prioritisation at city scale, - identify decision boundaries where rankings are fragile, - and show how these behaviours change under transfer.

The current phase has clear limits. Hazard is represented through a precipitation proxy rather than a hydrodynamic model. The outcome variable is synthetic and not calibrated against observed damage. Exposure is represented through a limited set of hydrography-based indicators. The results should therefore be interpreted as evidence about the behaviour of the probabilistic framework rather than as operational flood risk estimates.

Even with these limits, the project makes a clear shift:

from asking only “what is the estimated risk?” to asking “how reliable are the decisions based on that estimate?”

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